Known model relationships and held-driver responses

Linear

K = 4

Delay = 20 yr

No Y → X link

c = 0.38065 per 20-yr step

$g(X) = z$

Nonlinear

K = 4

Delay = 20 yr

No Y → X link

c = 0.38065 per 20-yr step

$g(X) = \tanh(2z)/0.797$

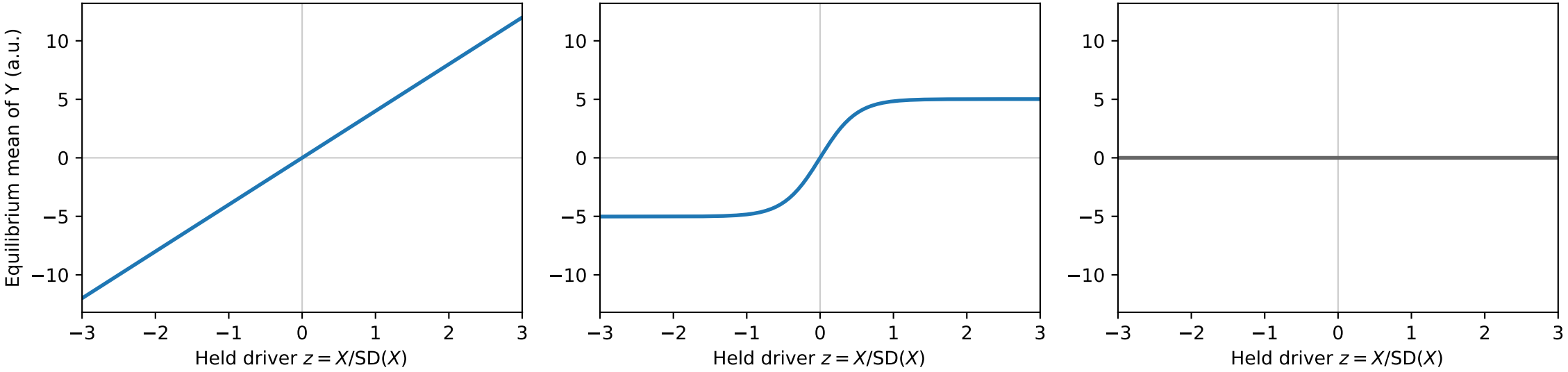
Independent

K = 0

No link in either direction

c = 0.00000 per 20-yr step

No forced response



$Y_t = a_Y Y_{t-1} + c\, g(X_{t-L}) + v_t$; $K = c/(1 - a_Y)$; held-driver mean $\overline{Y} = K\, g(X)$.

Independent, zero-mean process noise is present. Model K and causal-method scores have different meanings.